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Central Valley Regional Water Quality Control Board

3 May 2018

Ben Guzman, Jr.
San Joaquin Public Works
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Stockton, CA 95201

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NOTICE OF APPLICABILITY

GENERAL WASTE DISCHARGE REQUIREMENTS FOR SMALL DOMESTIC WASTEWATER TREATMENT SYSTEMS ORDER WQ 2014-0153-DWQ

FOR

SAN JOAQUIN COUNTY PUBLIC WORKS, CSA 44 ZONE G (LINNE) WWTP, SAN JOAQUIN COUNTY

San Joaquin County Public Works submitted a Report of Waste Discharge (RWD) dated July 2017 describing the CSA 44 Zone G (Linne) WWTP wastewater treatment plant (WWTP) in San Joaquin County. Additional information was submitted on 5 April 2018 via email. The WWTP provides treatment and disposal service for domestic wastewater generated from a residential subdivision. The WWTP discharge has been regulated by Waste Discharge Requirements (WDR) Order R5-2004-0078, which was adopted on 4 June 2004. Based on information provided in the RWD, the wastewater treatment system and discharge is consistent with the requirements of the State Water Resources Control Board (State Water Board) *General Waste Discharge Requirements for Small Domestic Wastewater Treatment Systems*, Order WQ 2014-0153-DWQ (General Order). This Notice of Applicability (NOA) serves as formal notice that upon rescission of Order R5-2004-0078 at an upcoming Board meeting, the discharge shall be regulated pursuant to the General Order and this NOA. You are hereby assigned Order WQ 2014-0153-DWQ-R5267 for the discharge. A copy of the General Order is enclosed and also available at:

http://www.waterboards.ca.gov/board_decisions/adopted_orders/water_quality/2014/wqo2014_0153_dwq.pdf

You should familiarize yourself with the entire General Order and its attachments, which describe mandatory discharge and monitoring requirements. The General Order contains operational and reporting requirements by wastewater system type. Sampling, monitoring, and reporting requirements applicable to your treatment and disposal methods must be completed in accordance with the appropriate treatment system sections of the General Order and the attached Monitoring and Reporting Program (MRP) 2014-0153-DWQ-R5267. The Discharger is responsible for all the applicable requirements that exist in the General Order and this NOA.

EXISTING FACILITY AND DISCHARGE DESCRIPTION

The CSA 44 Zone G (Linne) WWTP is owned by San Joaquin County Public Works (hereafter "Discharger") and operated by San Joaquin County Utilities Maintenance Division. The WWTP is located at 7580 West Bates Road in Tracy in San Joaquin County as shown on Attachment A, which is attached hereto and is made part of this NOA by reference. The WWTP serves 150 private residences and does not serve any commercial or industrial facilities. The site plan is shown on Attachment B, which is attached hereto and is made part of this NOA by reference.

Domestic wastewater is collected from 150 residences. Wastewater flows by gravity to a lift station through a bar screen into an Equalization Tank. The wastewater is then pumped into the anoxic basin at a reduced/equalized rate to minimize hydraulic and organic loading surges through the system. Flow through the entire activated sludge process is configured for gravity flow for continuous operation. The Anoxic Stage provides biological nitrogen removal (BNR) to the activated sludge system. Mixed liquor pumping brings mixed liquor solution, rich in nitrates, from the aeration basin to the anoxic basin for denitrification. Periodically, sludge is removed from the clarifier to the sludge tank. Once the sludge tank nears capacity, the sludge is dewatered. The free liquid is sent back to the anoxic basin and the sludge is sent off-site for disposal.

The tertiary treatment stage uses equalized flow to process secondary effluent through filtration and disinfection. Media filters are used to remove excess suspended matter from the secondary effluent before the effluent is disinfected by liquid chlorine solution. The filters are periodically backwashed and the water is returned to the equalization tank for re-processing. The effluent dosing tank is used to pump the final effluent to the dispersal field.

The 1.7 acre dispersal field consists of a pressure dosed trench and gravel system. The plant controls the dosing time and sequence of the dispersal zones (Zones 1 through 4). Sufficient land for a possible 100-percent replacement of the subsurface disposal system is available. Effluent samples are collected from the effluent tank, as shown on Attachment C. Wastewater quality and flow rates for 2016 and 2017 are shown below.

Effluent Quality ¹
(mg/L unless noted otherwise)

	pH	BOD	Nitrate as N	TKN	TDS	Total Coliform (MPN/100 ml)	Average Daily Flow (gallons)
2016	7.51	3.3	2.8	1	1,344	29	32,550
2017	7.43	0	2.1	2	1,238	5	33,178

¹ Average concentrations are shown.
BOD = biochemical oxygen demand
mg/L = milligrams per liter
MPN = most probable number
TKN = total Kjeldahl nitrogen
TDS = total dissolved solids

Five groundwater monitoring wells monitor groundwater quality up- and downgradient of the dispersal field. Monitoring well MW-4R and MW-5 are upgradient and MW-1 through MW-3 are located downgradient of the dispersal field. Depths to groundwater measured during the third quarter 2017 monitoring event ranged from 54 feet below the top of casing (btoc) to 57 feet btoc. Groundwater generally flows to the east. Groundwater monitoring data from first quarter 2017 sampling event is tabulated below:

Constituent	Upgradient Wells		Downgradient Wells			Concentrations Protective of Beneficial Use
	MW-4R	MW-5	MW-1	MW-2	MW-3	
TDS (mg/L)	1,190	1,590	1,540	1,560	1,250	450 ¹ to 1,500 ²
Nitrate Nitrogen (mg/L)	13.7	9	5	7	15.1	10 ³
Ammonia	<0.2	<0.2	<0.2	<0.2	<0.2	--
Total Coliform (MPN/100mL)	>1,600	<1.8	<1.8	<1.8	<1.8	<2
pH (standard units)	7.2	7.2	7.33	7.12	7.21	--
Concentrations in bold exceed a concentration protective of beneficial use.						
¹ Lowest agricultural water quality goal						
² Short-term Secondary Maximum Contaminant Level						
³ Primary Maximum Contaminant Level						
mg/L = milligrams per liter						
MPN = Most Probable Number						
TDS = Total Dissolved Solids						

Concentrations of nitrate as nitrogen and TDS in groundwater beneath the discharge area are relatively similar in upgradient and downgradient wells. However, concentrations of nitrate as nitrogen in the effluent are significantly lower than concentrations detected in groundwater. Because concentrations of TDS, coliform, and nitrate as nitrate in the upgradient wells exceed concentrations protective of beneficial use, it is likely that the groundwater beneath the dispersal field is being influenced by an upgradient source or existing poor quality groundwater in the area.

SITE-SPECIFIC REQUIREMENTS AND EFFLUENT LIMITS

Note that the General Order contains prohibitions and specifications that apply to all wastewater treatment systems as well as those that only apply to specific treatment and/or disposal systems. The specific requirements and effluent limits for your treatment system are summarized below.

The wastewater treatment operator must be certified and familiar with the requirements contained in the General Order, this NOA, and the MRP.

Requirements by Wastewater System Type, Section B of General Order

B.1 All Wastewater Systems

This section applies in its entirety to the CSA 44 Zone G (Linne) WWTP with the following site specific requirements.

B.1.a Effluent flow limits.

Treatment Unit	Flow Limit as Monthly Average
WWTP	55,000 gpd

B.1.1 Wastewater system setbacks.

Equipment or Activity	Domestic Well	Flowing Stream	Ephemeral Stream Drainage	Property Line	Lake or Reservoir
Septic Tank, Treatment System, & Collection System ¹	150 ft.	50 ft.	50 ft.	5 ft.	200 ft.
Leach Field ²	100 ft.	100 ft.	50 ft.	5 ft.	200 ft.

¹ Setbacks referenced under "Septic Tank, Aerobic Treatment Unit, Treatment System, or Collection System" in Table 3 of General Order.

² Setbacks referenced under "Leach Field" in Table 3 of General Order.

This is an existing facility, constructed prior to the issuance of the new General Order for Small Domestic Wastewater Treatment Systems, and the disposal area may not be in compliance with the setbacks included in the General Order. However, the WWTF will still be permitted under this General Order provided that nuisance conditions do not result from noncompliance. Expansion of a noncomplying wastewater treatment system shall trigger further evaluation of the setbacks, as described in Section B.1.I of the General Order.

B.3 Aerobic Treatment Units

The WWTP utilizes an aerobic treatment unit; therefore, this section applies in its entirety.

B.4 Activated Sludge Systems

The WWTP utilizes an activated sludge system; therefore, this section applies in its entirety.

B.6 Subsurface Disposal Systems

The WWTP utilizes a subsurface disposal system; therefore, this section applies in its entirety.

Effluent Limitations, Section D of General Order

This section applies in its entirety to the CSA 44 Zone G (Linne) WWTP and shall include the following site specific limitations.

Effluent Limitations

The following limits apply to effluent from WWTP.

Constituent	Units	Limit
BOD	mg/L	40
Total Nitrogen	mg/L	10 mg/L

Effluent Limit Rationale

The treatment system is subject to technology performance effluent limits for biochemical oxygen demand (BOD) as specified in the General Order.

Staff evaluated the need for a total nitrogen effluent limit using the method contained in the General Order and determined that a nitrogen effluent limit is required because the monthly average flow will be less than 20,000 gpd.

Technical Report Preparation Requirements, Provisions Section E.1 of General Order

The following technical reports shall be submitted as described below:

1. By **1 October 2018**, the Discharger shall submit a *Spill Prevention and Emergency Response Plan* (Response Plan) consistent with the requirements of General Order Provision E.1.a.
2. By **1 October 2018**, the Discharger shall submit a *Sampling and Analysis Plan* consistent with the requirements of General Order Provision E.1.b.
3. By **1 October 2018**, the Discharger shall submit a Sludge Management Plan consistent with the requirements of General Order Provision E.1.c.

MONITORING AND REPORTING PROGRAM

The Discharger shall comply with MRP 2014-0153-DWQ-R5267, which is attached hereto and made part of this NOA by reference.

ENFORCEMENT

Please review this NOA carefully to ensure that it completely and accurately reflects the discharge. Discharge of wastes other than those described in this NOA is prohibited. Prior to allowing changes to the wastewater strength or generation rate, or to the method of waste disposal, you must contact the Central Valley Regional Water Board to determine if submittal of an RWD is required.

San Joaquin County Public Works and San Joaquin County Utilities Maintenance Division will generate the waste subject to the terms and conditions of WQ 2014-0153-DWQ-R5267 and will maintain exclusive control over the discharge. As such, San Joaquin County Public Works and San Joaquin County Utilities Maintenance Division are primarily responsible for compliance with this NOA, MRP, and General Order, with all attachments. Failure to comply with the requirements in the General Order or this NOA could result in an enforcement action as authorized by provisions of the California Water Code.

ANNUAL FEES

Staff has determined the discharge is a threat to water quality and complexity rating of 2-B. The annual fee corresponding to a threat to water quality and complexity of 2-B is currently \$14,929. The fee is due and payable on an annual basis until coverage under the General Order is formally rescinded. Please note that the annual fees are reviewed each year and may change. If the wastewater discharge ceases, you must provide written notice so that we can terminate coverage under the General Order and no longer bill you.

DOCUMENT SUBMITTAL

All monitoring reports and other correspondence should be converted to searchable Portable Document Format (PDF) and submitted electronically. Documents that are less than 50 MB should be emailed to:
centralvalleysacramento@waterboards.ca.gov.

To ensure that your submittal is routed to the appropriate staff person, the following information should be included in the body of the email or any documentation submitted to the mailing address for this office:

Facility Name: CSA 44 Zone G (Linne) WWTP, San Joaquin County		
Program: Non-15 Compliance	Order: 2014-0153-DWQ-R5267	CIWQS Place ID: 237113

Documents that are 50 MB or larger should be transferred to a CD, DVD, or flash drive and mailed to:

Central Valley Regional Water Quality Control Board
ECM Mailroom
11020 Sun Center Drive, Suite 200
Rancho Cordova, CA 95670

Now that the Notice of Applicability has been issued, the Board's Compliance and Enforcement section will take over management of your case. Brendan Kenny is your new point of contact for any questions about the General Order. If you find it necessary to make a change to your permitted operations, Brendan will direct you to the appropriate Permitting staff. You may contact Brenda at (916) 464-4635 or at bkenny@waterboards.ca.gov.

For 
Pamela C. Creedon
Executive Officer

enc: Water Quality Order WQ 2014-0153-DWQ
Monitoring and Reporting Program 2014-0153-DWQ-R5267
Attachment A, Site Location Map
Attachment B, Site Plan
Attachment C, Wastewater Treatment System Schematic

cc w/out enc: Timothy O'Brien, State Water Resources Control Board, Sacramento
San Joaquin County Environmental Health Department, Stockton